
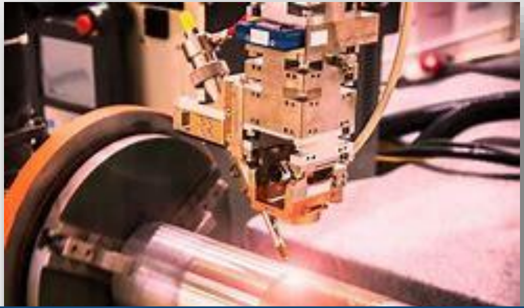


Chapter 7

➢ Inductive Sensors



Contents

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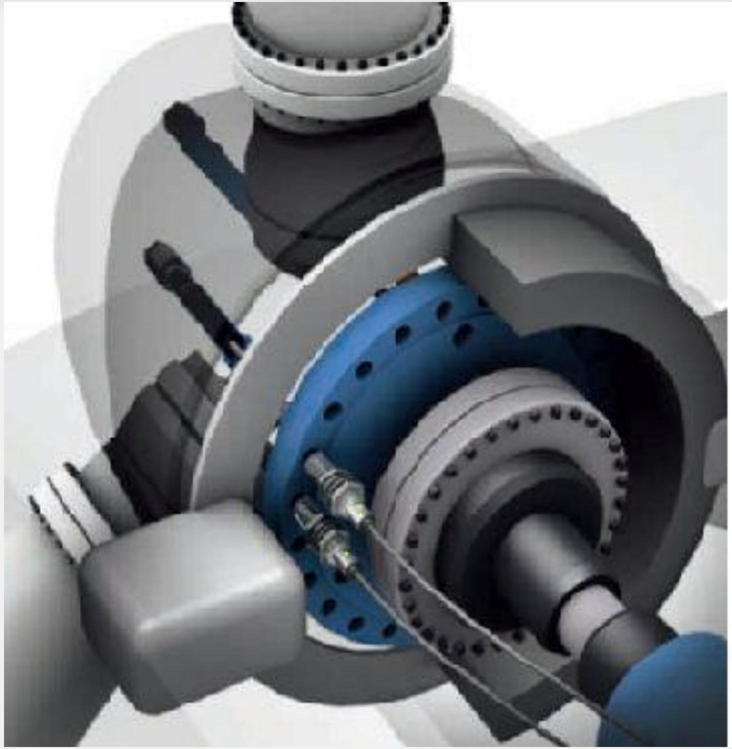
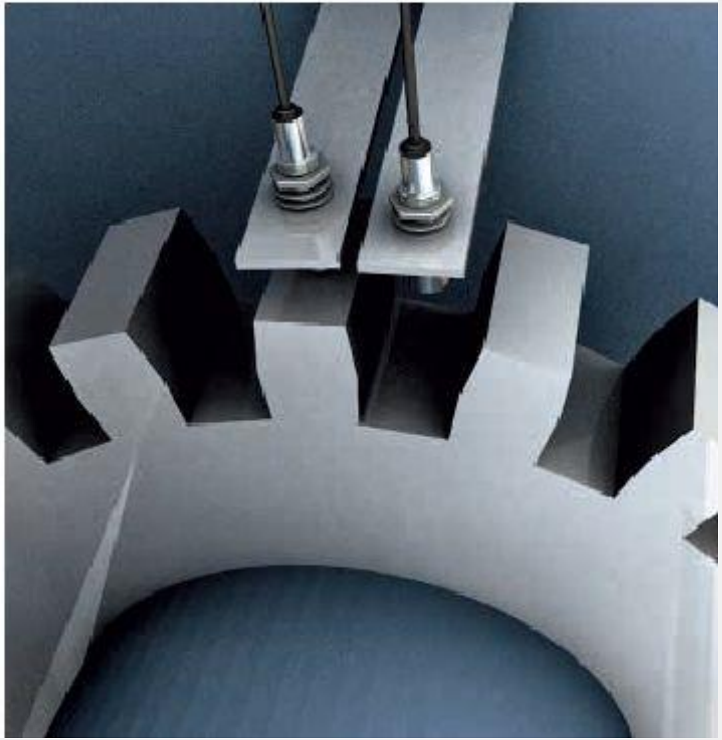
Practical engineering problem

Inductive Sensors

汽车制造业生产过程中，大量使用电感式传感器作为定位检测，以满足汽车制造工艺要求

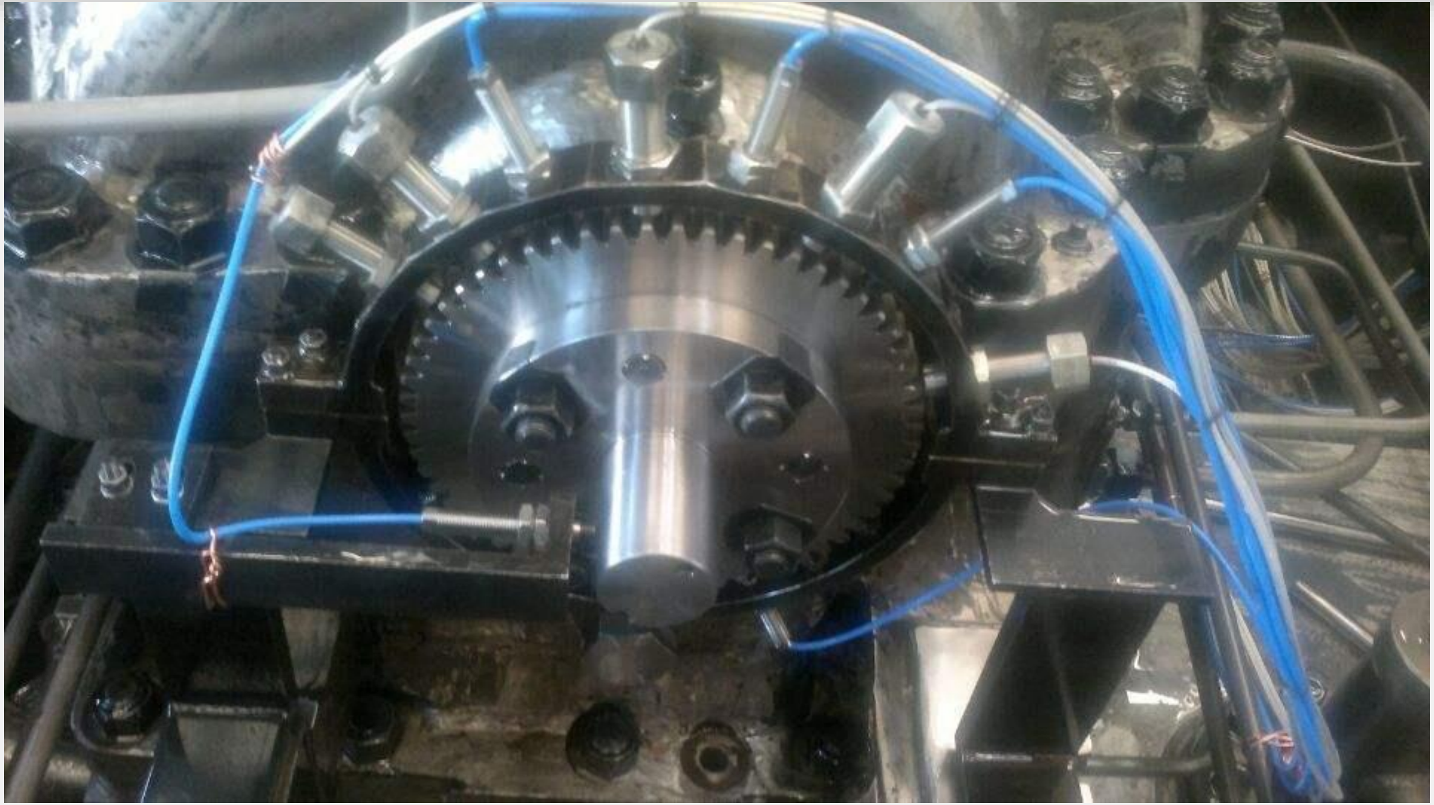
Inductive Sensors





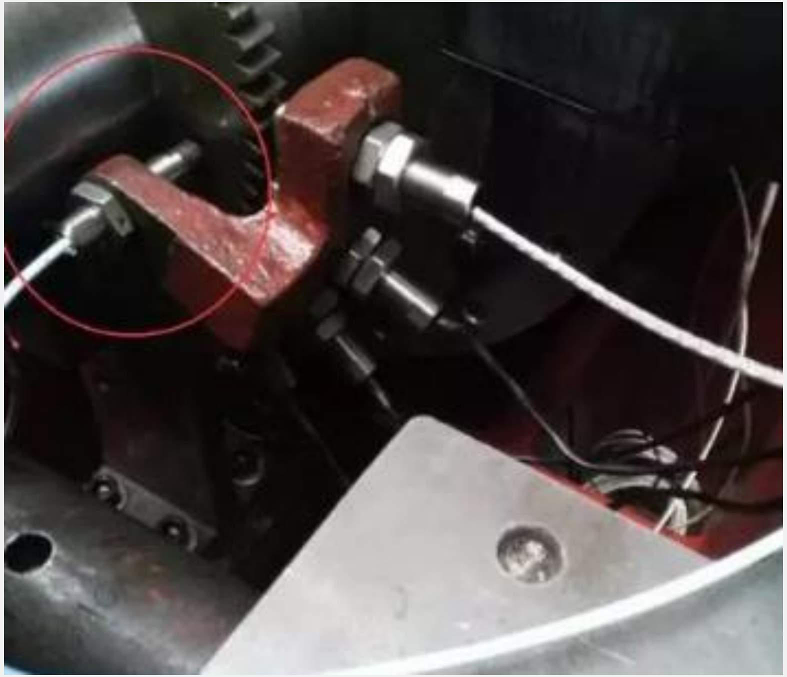
风力发电机振动监测

Inductive Sensors



电涡流传感器用于旋转机械（转轴）的振动监测

Inductive Sensors



电涡流传感器用于振动监测

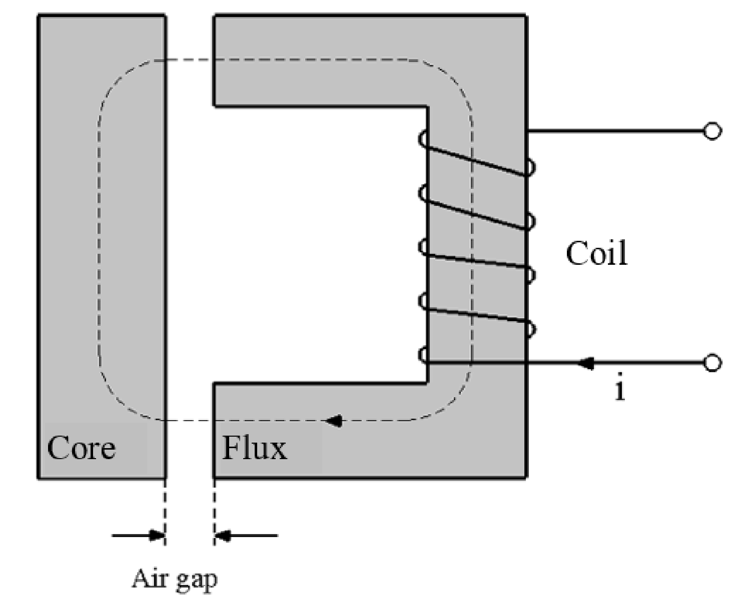
2

Principle

Inductive Sensors

- The self-inductance inductance sensor
- The mutual inductance sensor
- The eddy current inductance sensor

7.2 Self-Inductance Inductive Sensor



The magnetomotive force （mmf）：

$$mmf = Flux \times Reluctance = \Phi \times R$$

The magnetic flux Φ：

$$\Phi = \frac{ni}{R}$$

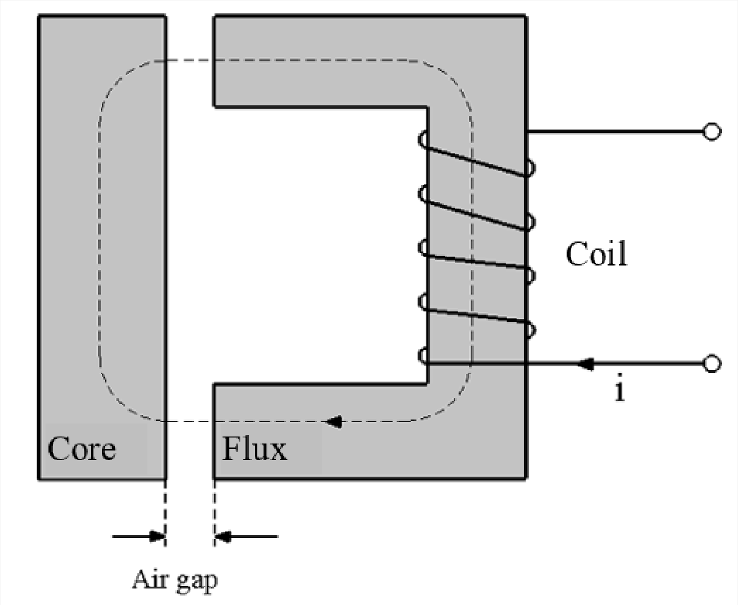
The total flux

$$\Psi = n\Phi = \frac{n^2 i}{R}$$

The self-inductance *L* of the coil：

$$L = \frac{\Psi}{i} = \frac{n^2}{R}$$

7.2 Self-Inductance Inductive Sensor

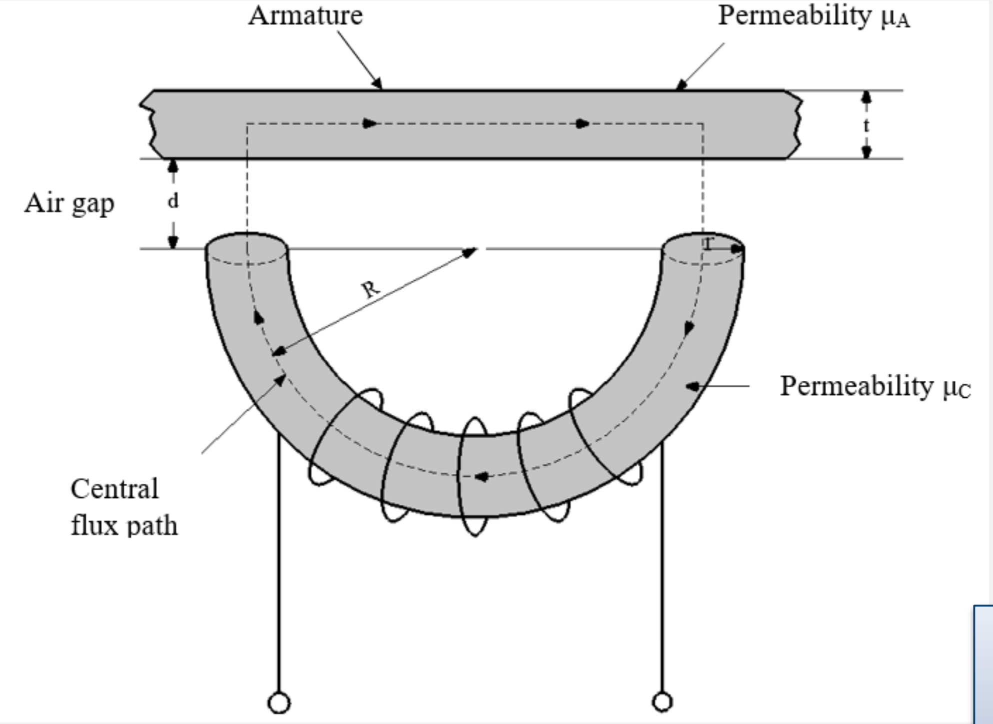


The reluctance R:

$$R = \frac{l}{\mu\mu_0 A}$$

Where l is the total length of the flux path, μ is the relative permeability of the magnetic circuit material, μ_0 is the permeability of free space ($= 4\pi \times 10^{-7}$H/m), A is the cross-sectional area of the flux path.

7.2.1 The Single-Coil Linear Variable-Reluctance Sensor



Sum of the individual reluctances:

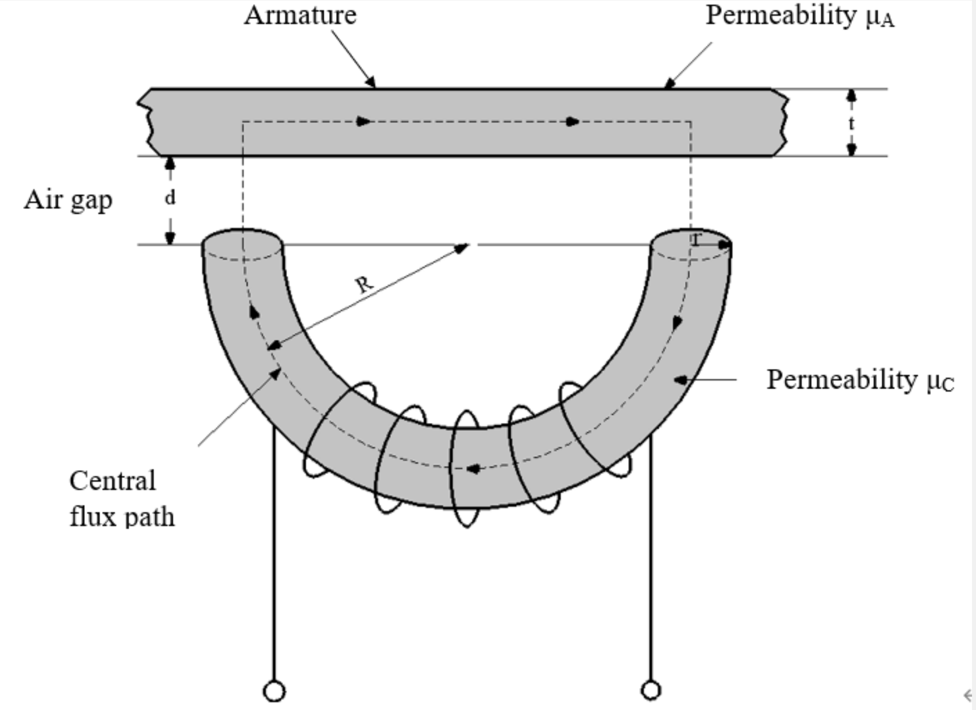
$$R_T = R_C + R_G + R_A$$

Where R_C, R_G and R_A are the reluctances of the core, air gap, and armature, respectively.

$$R_T = \frac{R}{\mu_C \mu_0 r^2} + \frac{2d}{\mu_0 \pi r^2} + \frac{R}{\mu_A \mu_0 r t}$$

In this equation, all the parameters are fixed except for the one independent variable — the air gap

7.2.1 The Single-Coil Linear Variable-Reluctance Sensor



$$L = \frac{\Psi}{i} = \frac{n^2}{R}$$

$$R_T = R_0 + kd$$

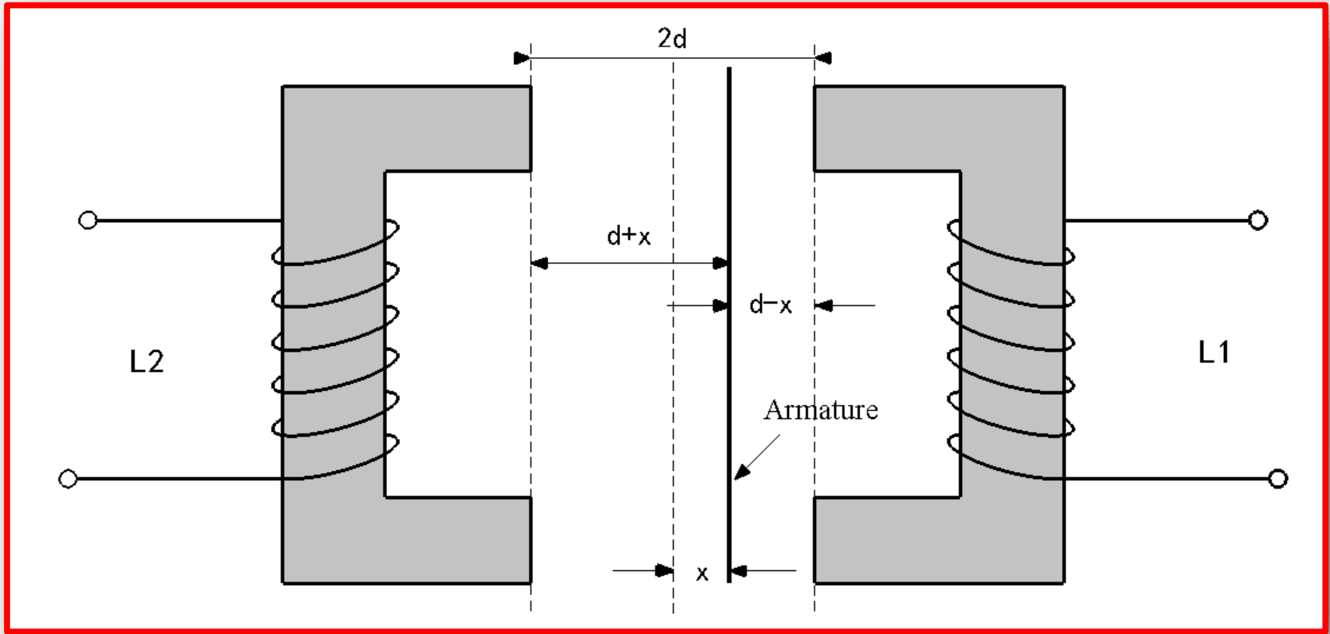
$$L = \frac{n^2}{(R_0 + kd)} = \frac{L_0}{(1 + \alpha d)}$$

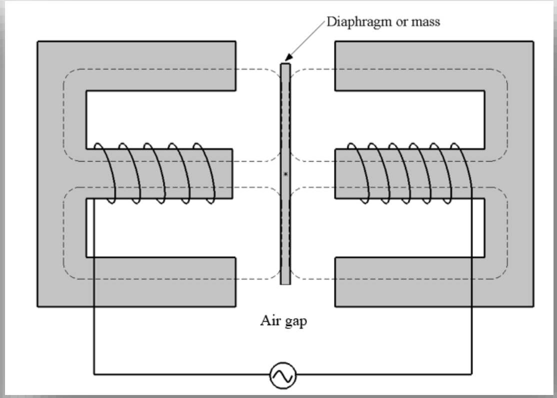
where L_0 is the inductance at zero air gap, and $\alpha = k/R_0$.

It can be seen from Equation 7.9 that the relationship between L and α is nonlinear.

Inductive Sensors

7.2.2 The Variable-Differential Reluctance Sensor



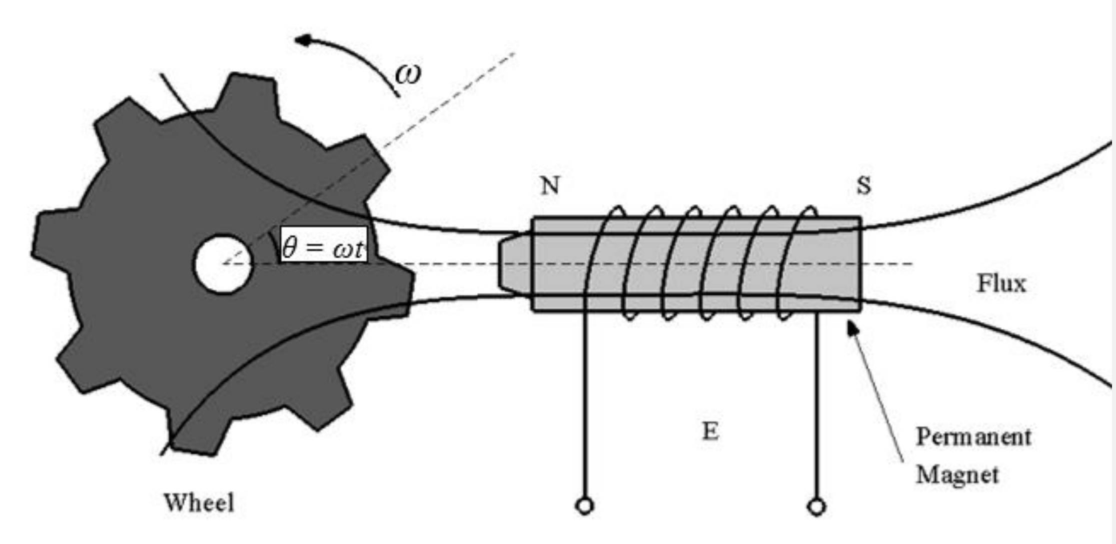


$$L_1 = \frac{L_{01}}{[1 + \alpha(d - x)]}$$

$$L_2 = \frac{L_{02}}{[1 + \alpha(d - x)]}$$

Inductive Sensors

7.2.3 Variable-Reluctance Tachogenerators



$$\Psi(\theta) = A + B\cos m\theta$$

$$E = -\left[\frac{\mathrm{d}\Psi(\theta)}{\mathrm{d}t}\right] = -\left[\frac{\mathrm{d}\Psi(\theta)}{\mathrm{d}\theta}\right] \times \left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)$$

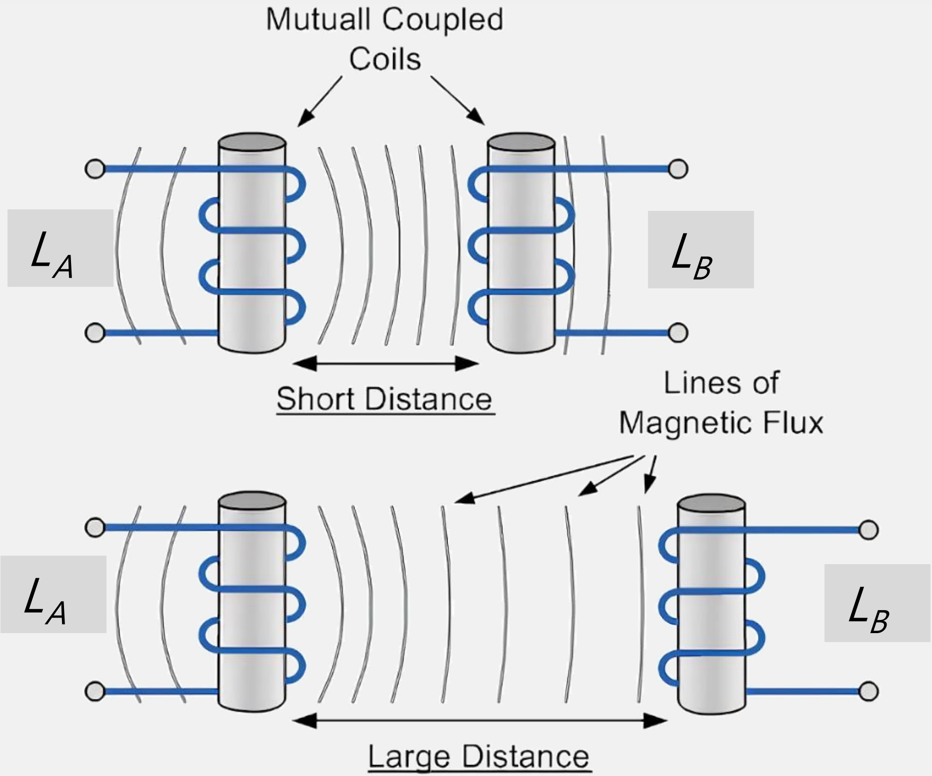
$$E = Bm\omega \sin m\omega t$$

Inductive Sensors

7.3 Mutual Inductance Sensor: LVDT

The mutual inductance of the coils is given by

$$M = K\sqrt{L_A L_B}$$



7.3 Mutual Inductance Sensor: LVDT

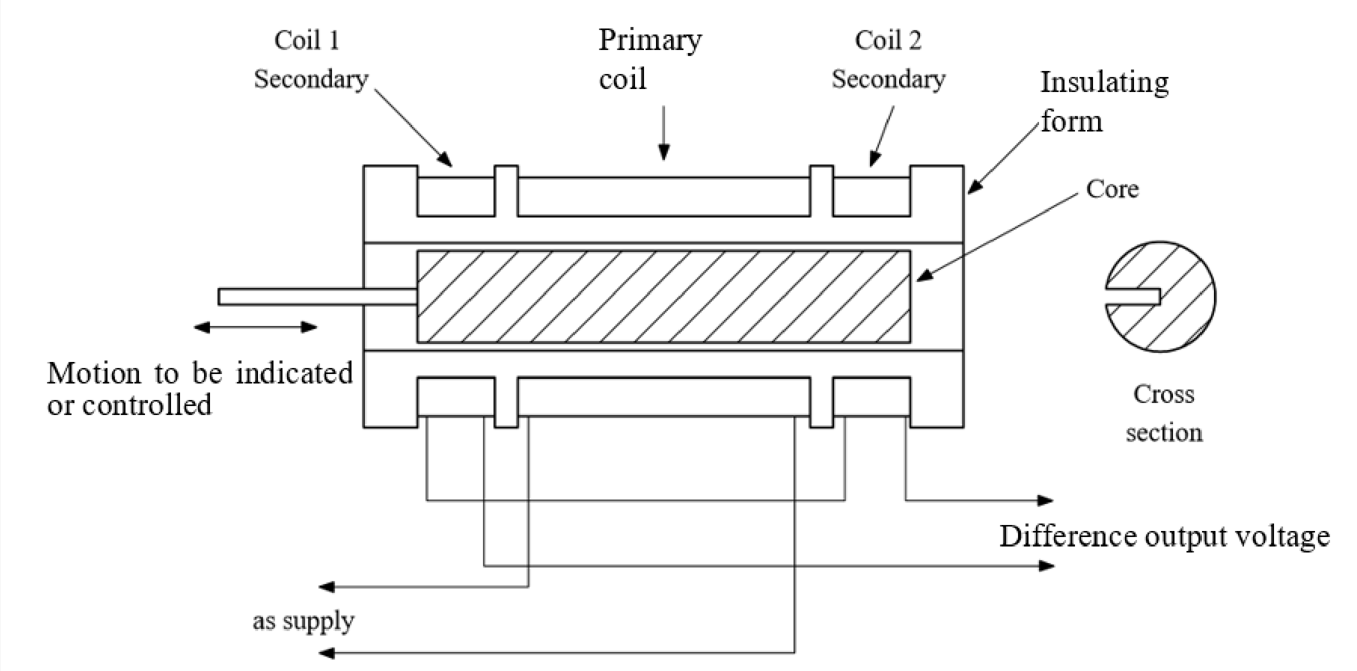
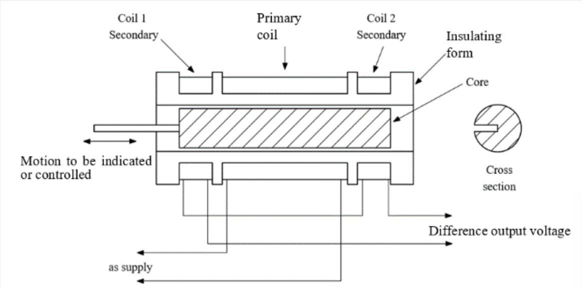
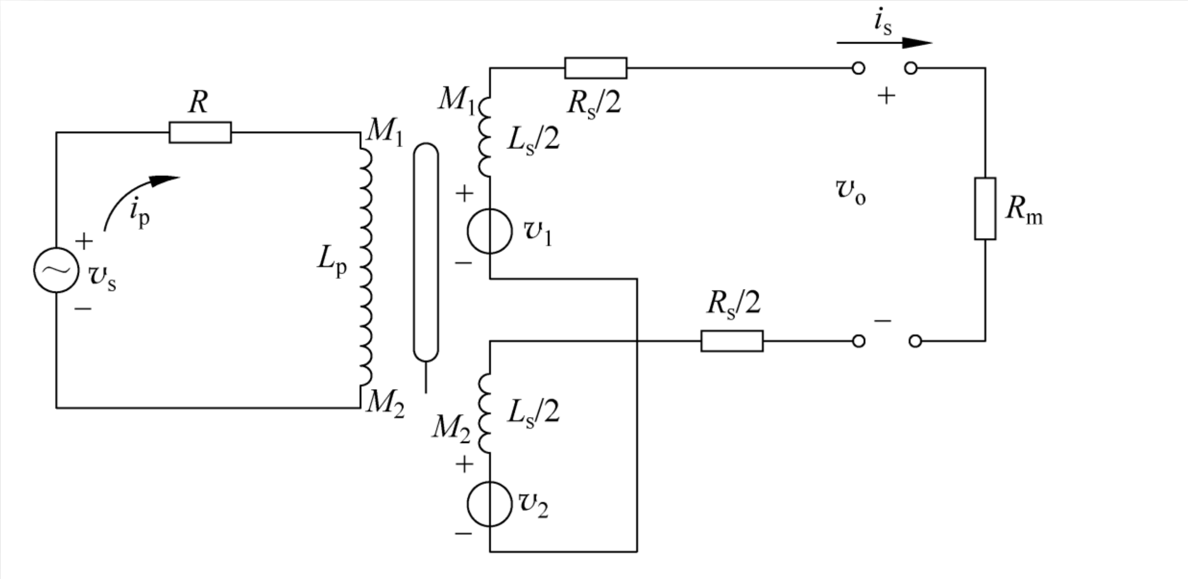


Figure 7.5 The linear-Variable-Differential-Transformer (LVDT)

Inductive Sensors

7.3 Mutual Inductance Sensor: LVDT





Inductive Sensors

7.3 Mutual Inductance Sensor: LVDT

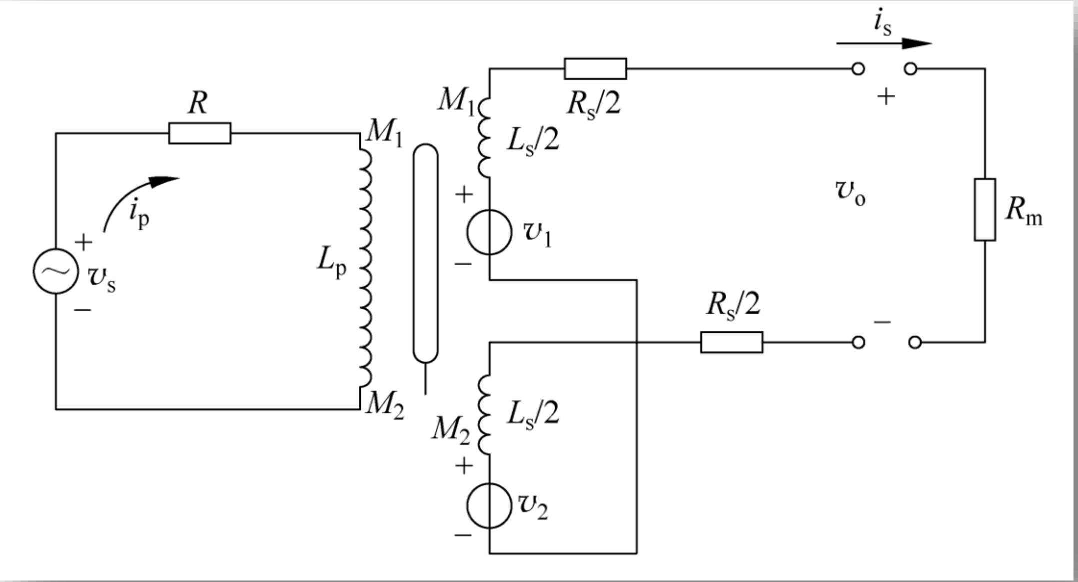
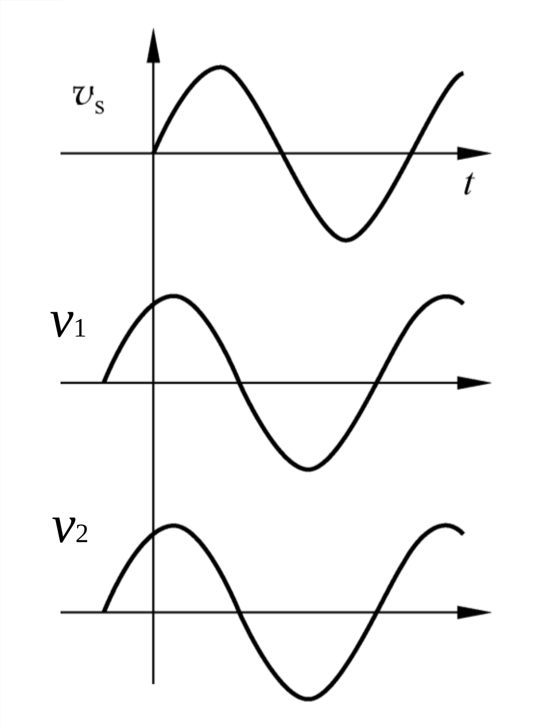
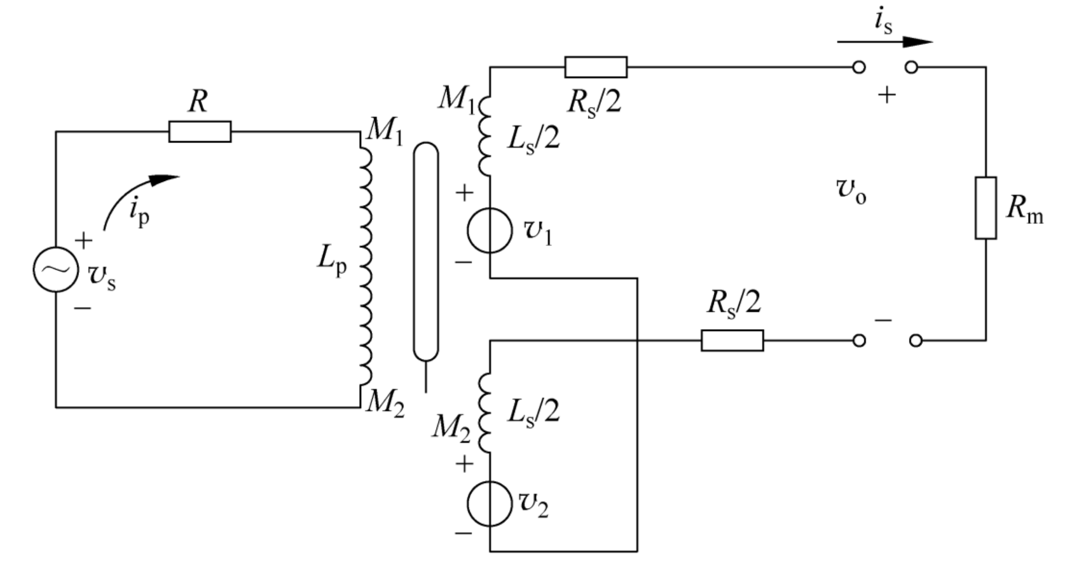


Figure 7.6 Voltages Induced in the Secondaries of a Linear-Variable Differential-Transformer



Inductive Sensors

7.3 Mutual Inductance Sensor: LVDT



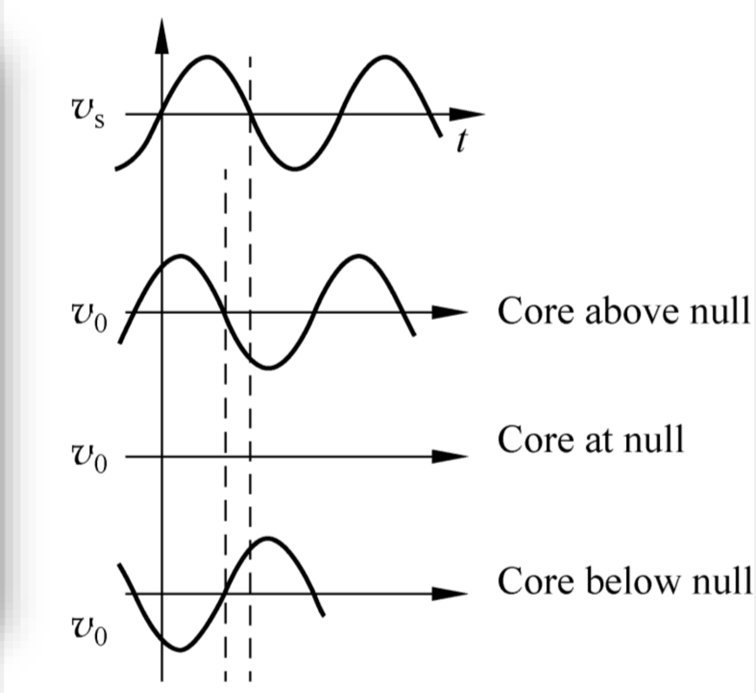


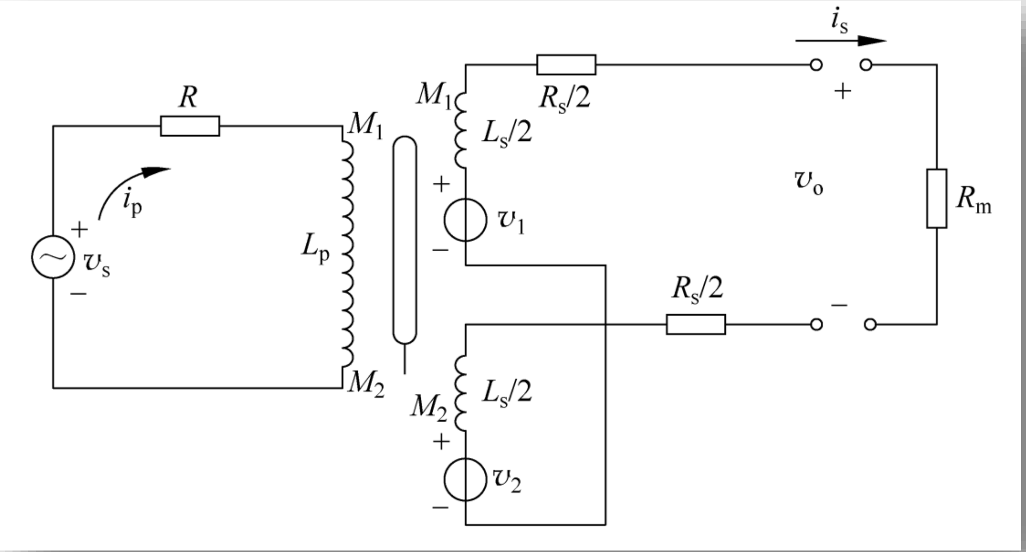
Figure 7.6 Voltages Induced in the Secondaries of a Linear-Variable Differential-Transformer

Inductive Sensors

7.3 Mutual Inductance Sensor: LVDT

$$v_1 = M_1 s i_p \text{ and } v_2 = M_2 s i_p$$

$$v_0 = v_1 - v_2 = (M_1 - M_2) s i_p$$



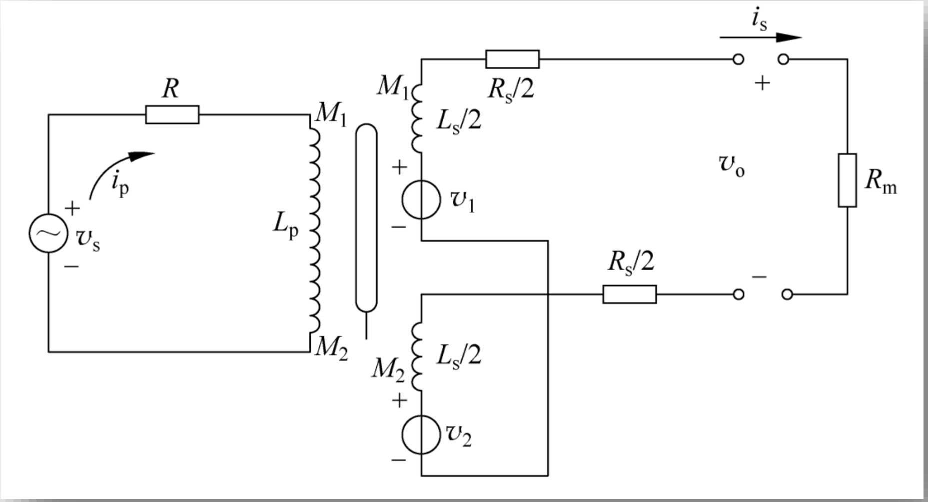
$$v_s = i_p(R + sL_p)$$

$$\frac{v_0}{v_s} = \frac{(M_1 - M_2)s}{(R + sL_p)}$$

Inductive Sensors

7.3 Mutual Inductance Sensor: LVDT

However, if there is a current due to output signal processing, then describing equations may be modified as:



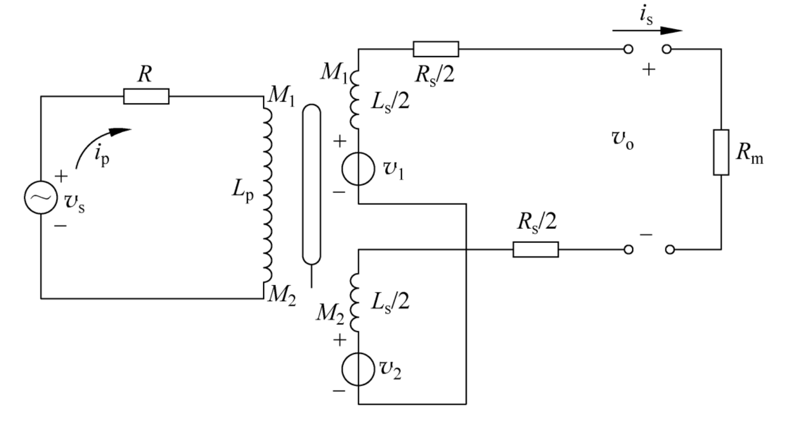
$$v_0 = R_m\, i_s$$

$$v_s = i_p\left(R + sL_p\right) - \left(M_1 - M_2\right) s i_s$$

$$\boxed{\frac{v_0}{v_s} = \frac{R_m(M_1 - M_2)s}{\{[L_S L_p - (M_1 - M_2)^2]s^2 + \left[L_p(R + R_m) + RL_S\right]s + (R_s + R_m)R\}}}$$

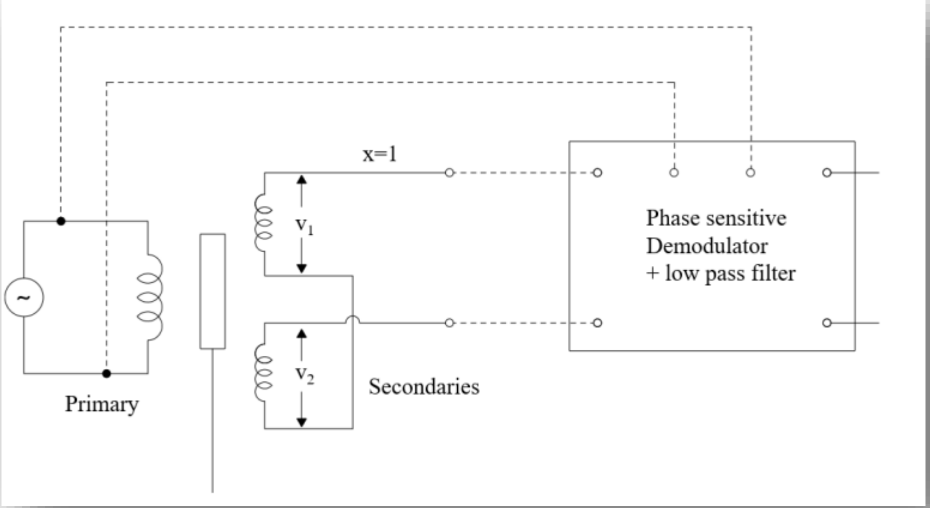
7.3 Mutual Inductance Sensor: LVDT

This is a second-order system, which indicates that due to the effect of the numerator of Equation 7.21, the phase angle of the system changes from +90° at low frequencies to –90° at high frequencies. In practical applications, the supply frequency is selected such that at the null position of the core, the phase angle of the system is 0° .



$$\frac{v_0}{v_s} = \frac{R_m(M_1 - M_2)s}{\{[L_S L_p - (M_1 - M_2)^2]s^2 + \left[L_p(R + R_m) + RL_S\right]s + (R_s + R_m)R\}}$$

7.3 Mutual Inductance Sensor: LVDT

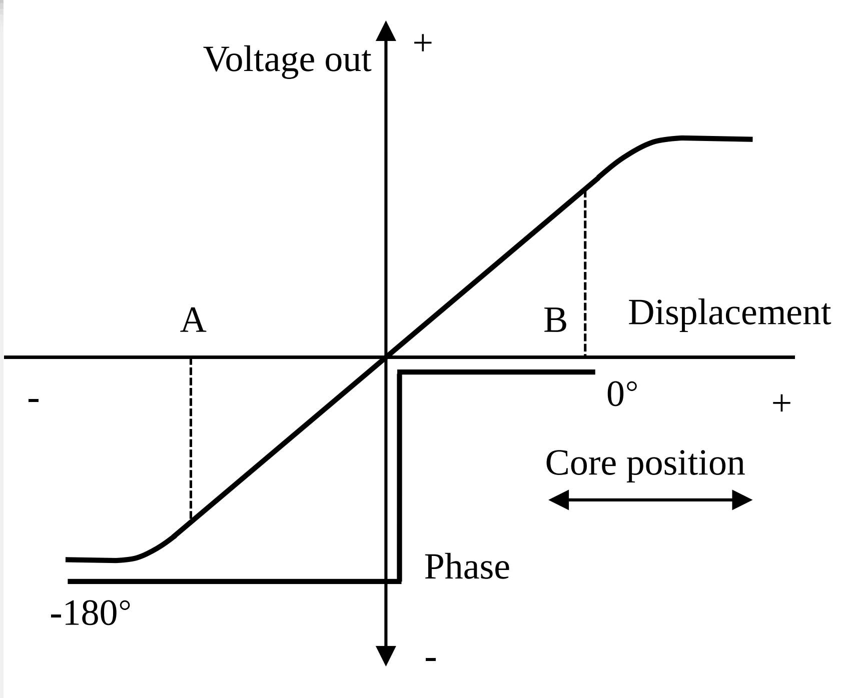


Phase-sensitive（相敏）demodulator (解调器) and filter arrangement is commonly used, as shown in Fig.7.7 (a). Phase-sensitive demodulator is commonly used to obtain displacement-proportional signals from LVDTs and other differential type inductive sensors. They convert the AC outputs from the sensors into DC values and also indicate the direction of movement of the core from the null position.

How to determine the direction of motion of a moving object?

7.3 Mutual Inductance Sensor: LVDT

A typical output of the phase-sensitive demodulator is shown in Fig.7.7 (b). The relationship between output voltage v_o and phase angle is also shown against core position x.



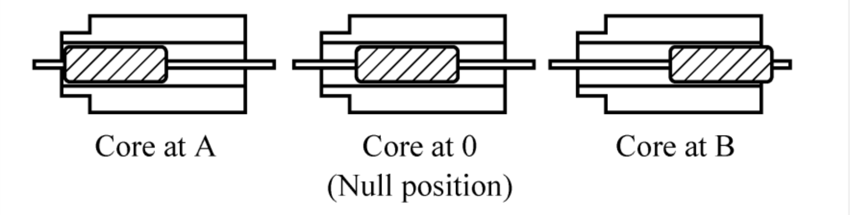
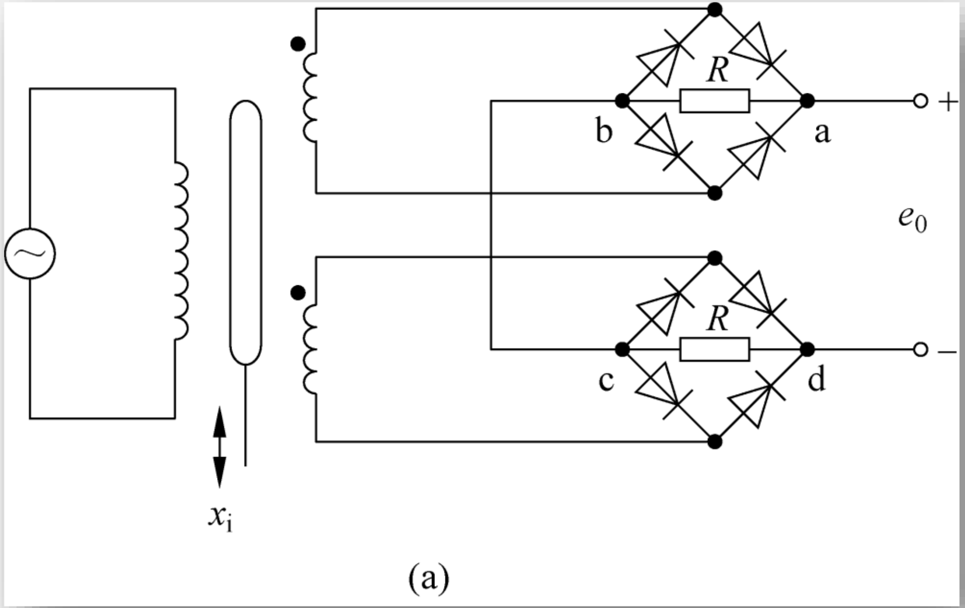


Figure 7.7 The Phase-Sensitive Demodulator

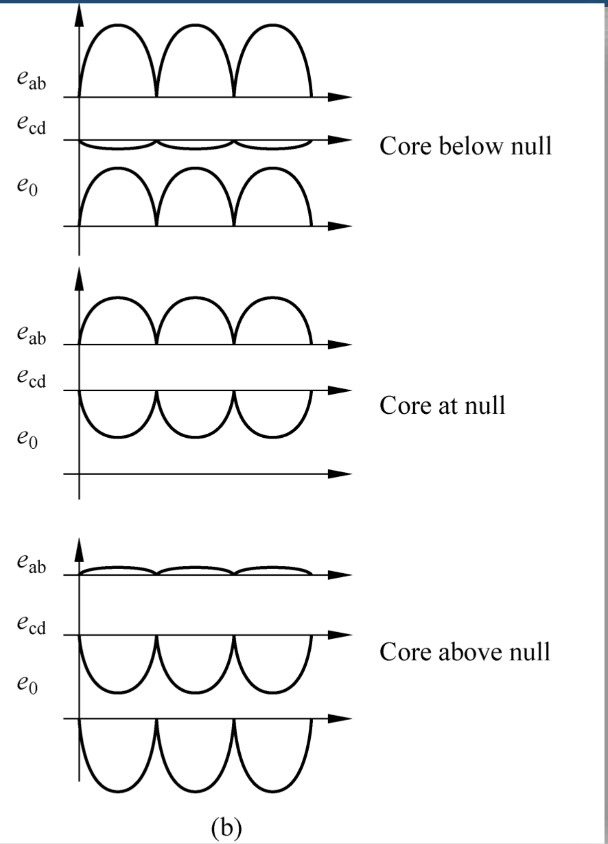
Inductive Sensors



A typical phase-sensitive demodulation circuit maybe constructed, based on diodes(二极管) shown in Fig. 7.8(a). This arrangement is useful for very slow displacements, usually less than 1 or 2 Hz. In this figure, bridge 1 acts as a rectification（整流） circuit for secondary 1, and bridge 2 acts as a rectifier（整流器） for secondary 2.

Figure 7.8 Typical phase-sensitive demodulation circuit based on diode

Inductive Sensors

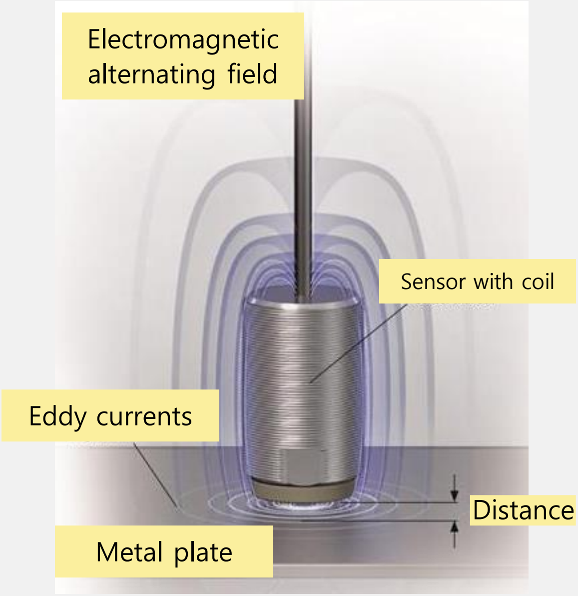


The net output voltage is the difference between the outputs of two bridges, as in Fig.7.8(b). The position of the core can be determined from the amplitude of the DC output, and the direction of the movement of the core can be determined from the polarity(极性) of the DC voltage. For rapid movements of the core, the outputs of the diode bridges need to be filtered, where in only the frequencies of the movement of the core pass through and all the other frequencies produced by the modulation process are filtered.

Figure 7.8 Typical phase-sensitive demodulation circuit based on diode

7.4 Eddy Current (电涡流)

An eddy or Foucault current is an electromagnetic phenomenon discovered by the French physicist Leon Foucault in 1851. When a conductor is exposed to a changing magnetic field (e.g., created by an AC coil, as shown in Fig.7.9), a circulating flow of electrons (eddy current) is induced in the conductor.



Inductive Sensors

7.4 Eddy Current

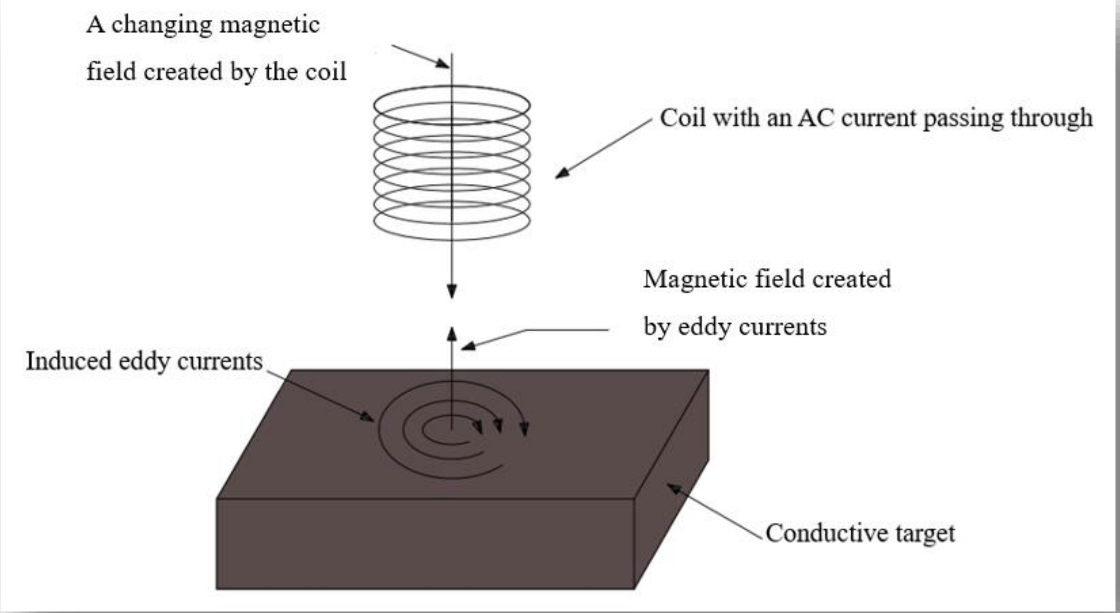


Figure 7.9 Eddy-Current Principle

The generation of these eddy currents takes energy from the coil and appears as an increase in the electrical resistance of the coil. The eddy currents also generate their own magnetic fields that oppose the magnetic field of the coil, and thus change the inductive reactance（电抗） of the coil.

Inductive Sensors

7.4 Eddy Current

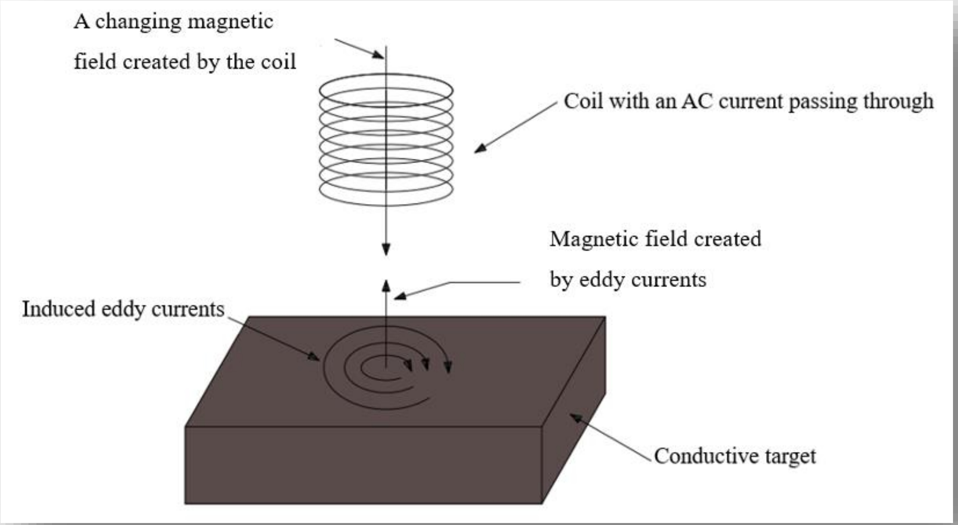


Figure 7.9 Eddy-Current Principle

Resistance and inductive reactance vectors add up to the total impedance of the coil. This impedance increase can be measured with an eddy-current probe, from which much information about the test material can be obtained. Typical eddy-current testing devices are designed to measure these energy losses. The eddy-current loss, P_E (in watts), is measured by

$$P_E = K_E B_{\max}^2 f^2 t_h^2 V_{\mathrm{vol}}$$

Inductive Sensors

If a conductor is carrying a high AC, the distribution of the current is not evenly dispersed throughout the cross section of the conductor. This is due to two independent effects known as skin effect and proximity effect. Skin effect often occurs in a single conductor (i.e., no other conductors are nearby), in which the current density near the surface of the conductor is the largest and the current density decreases as the depth increases. For an eddy current case the skin effect suggests that the induced eddy current flows along the "skin" of the conductor.

集肤效应和邻近效应

Proximity effect occurs where two or more conductors are close to each other. Refer to Fig. 7.10, if each conductor carries a current flowing in the same direction, the areas of the conductors in close proximity experience more magnetic flux than the remote areas. Consequently, current distribution is not even throughout the cross section, and a greater proportion of current is carried by the remote areas. This phenomenon can also

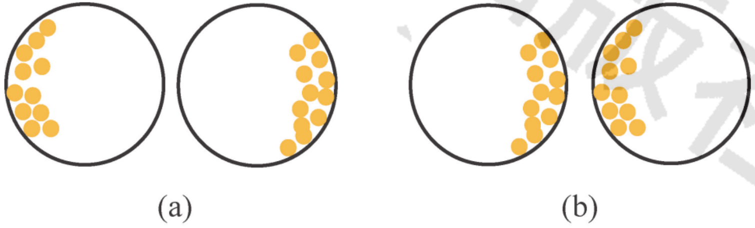


Figure 7.10 Proximity effect for two wires with same or different current direction

(a) Two wires with the same current direction; (b) Two wires with the opposite current direction

be explained by Lorentz force that pushes the charge carries (e.g., electrons or holes) and causes the higher charge densities in the remote areas. If the currents are in opposite directions, the areas in close proximity will carry the greater density of current.

Inductive Sensors

7.5 Advantages/Disadvantages of Inductive Sensor

The advantages of the inductive sensor include the following.

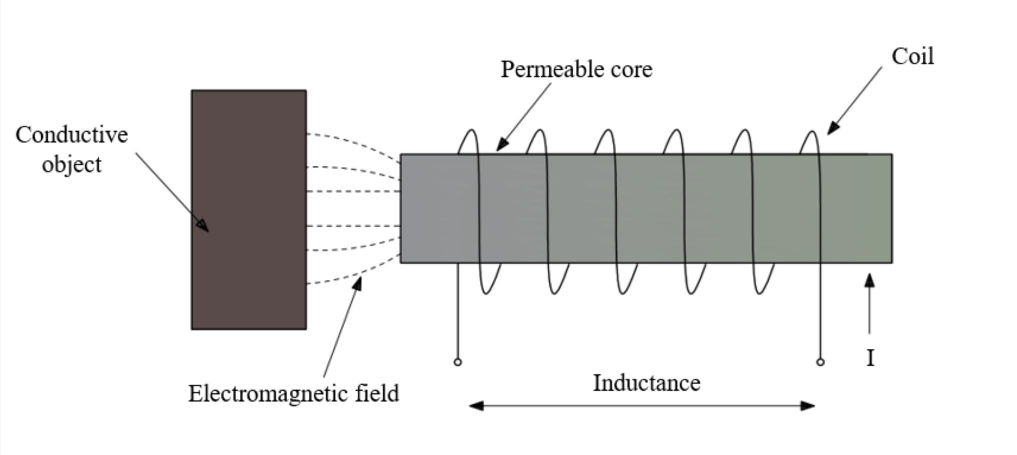
- The inductive sensors can work in any environmental conditions like humidity and high temperatures. These can give high performance in the industrial environment also.
- These have high accuracy and stable operating range with good life-span.
- These can be operated in high switching rates in industrial applications.
- These type sensors can be operated in wide ranges used in various applications.

The disadvantages of the inductive sensor include the following.

- The working and operating range of inductive sensor depends on the construction and temperature conditions.
- It depends on the magnetic field of the coil.

Inductive Sensors

7.6 Applications of the Inductive Sensor

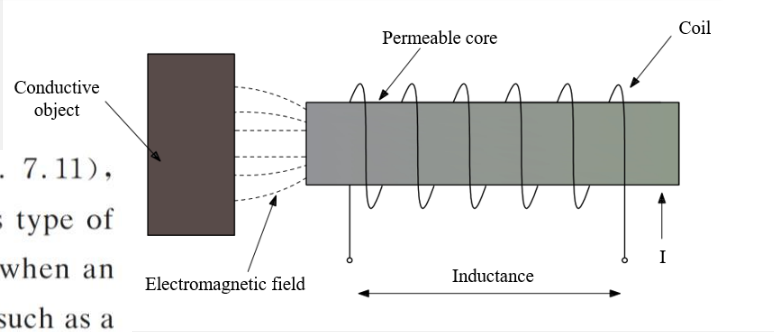


7.6.1 Inductive Proximity Switch

Inductive Sensors

7.6 Applications of the Inductive Sensor

7.6.1 Inductive Proximity Switch



One of the common inductive sensors is an inductive proximity switch (see Fig. 7.11), primarily used to detect the presence of a conductive object in front of it. This type of sensor has a coil wound around an iron core, creating an electromagnetic field when an oscillating current flowing through the coil. If a magnetic or conductive object, such as a metal plate, is placed within the magnetic field around the sensor, the inductance of the coil changes. The sensor's detection circuit detects this change and produces an output voltage to trigger the switch ON. Inductive proximity sensors are widely used for traffic light control at intersections. Rectangular inductive loops of wire are buried into the road surface. When a vehicle passes over the loop, the metallic body of the vehicle changes the loop's inductance and triggers the circuit to change the traffic lights. One disadvantage of this type of sensors is that they are "omnidirectional," meaning that they will sense a metallic object above, below, or to the side of it. Also, they cannot detect nonmetallic objects (capacitive sensors can detect nonmetallic objects with a typical sensing range of 0.1-12 mm)[29].

Inductive Sensors

7.6 Applications of the Inductive Sensor

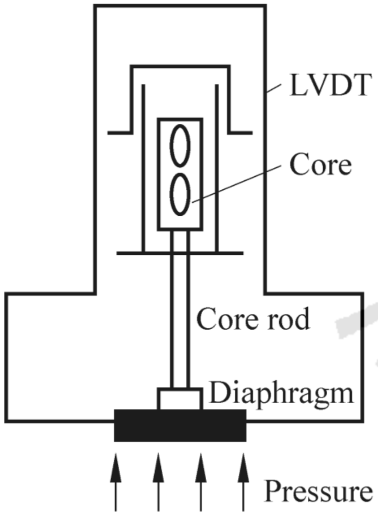


Figure 7.12 LVDT pressure sensor

7.6.2 LVDT pressure sensor

To use an LVDT sensor to measure pressure, the pressure must be converted to a linear displacement. Any pressure-sensitive element that changes size or shape as a pressure is applied, such as bellows, Bourdon tubes, or aneroid capsules, can be used. As a pressure is applied, the core is displaced from its null position, causing an output voltage. Fig. 7.12 shows an LVDT pressure sensor in which a pressure diaphragm drives the core up and down and varies the inductive coupling between the primary and secondary windings. The LVDT is a differential device and can be made to measure absolute, relative, or differential pressures.[29]

7.6.3 Flaw detection sensors

There are several designs for inductive sensing probes. Fig. 7.13 shows the simplest design. It contains a coil, a plastic tip, a stainless-steel housing, and a cable. Tips are usually made of dielectric materials (e. g. plastic or epoxy). The sensing coil is encapsulated in the tip and installed in a stainless-steel housing.[29]

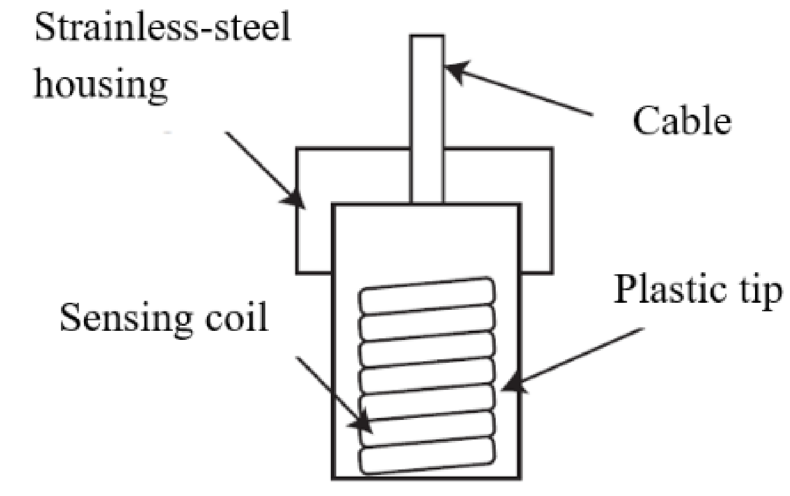


Figure 7.13 The Internal Structure of a Simple Probe

Inductive Sensors

Some probes have two coils in one tip: one is the excitation coil and the other is a receiving coil (see Fig. 7.14). The former creates an alternating magnetic field to induce eddy currents in a conductive target, while the latter picks up the change of the coil magnetic field due to its interaction with the induced eddy-current magnetic fields. The latter is placed in the very front of the probe where the strength of the induced field is higher. The coil former can be made of machinable ceramic. The coils may be fiberglass-insulated copper wires, encapsulated in a ceramic tube filled with ceramic

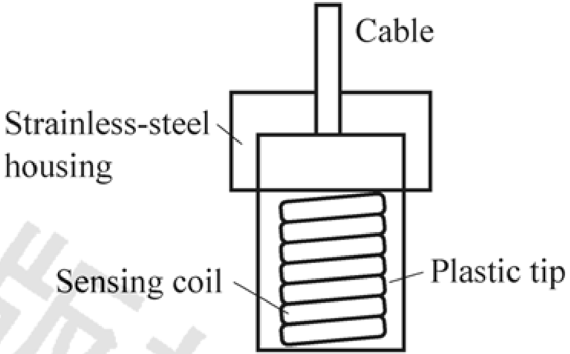


Figure 7.13 The internal structure of a simple probe

Inductive Sensors

Probe bodies and coils are available in a large variety of shapes and sizes, and can be custom-designed as well. Fig. 7.15 shows the tangential and pancake types. Tangential coils cannot detect certain cracks effectively. For example, circumference cracks on a tubular target because the cracks and the flow of the induced eddy current have the same direction, while pancake coil, however, can ensure that the circumference cracks be interrupted by the eddy-current flow (see Fig. 7.16).

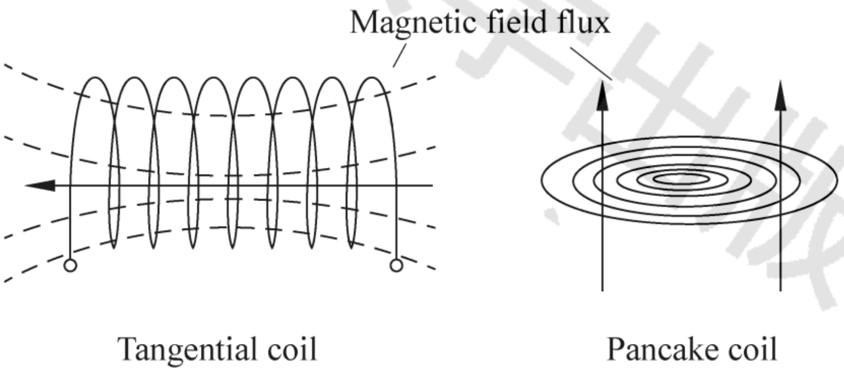
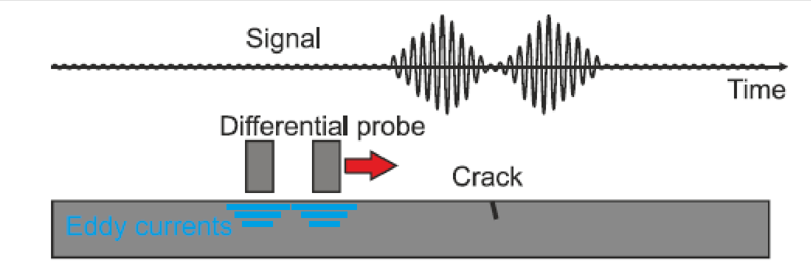


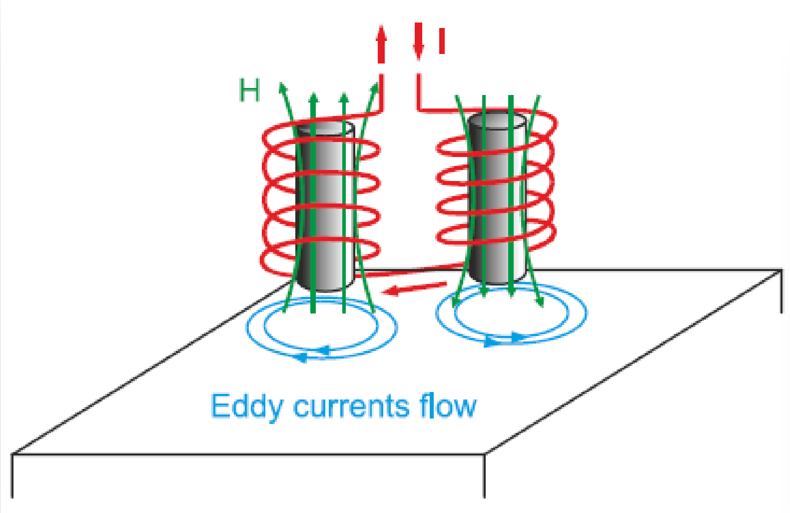
Figure 7.15 Tangential and pancake coil configurations

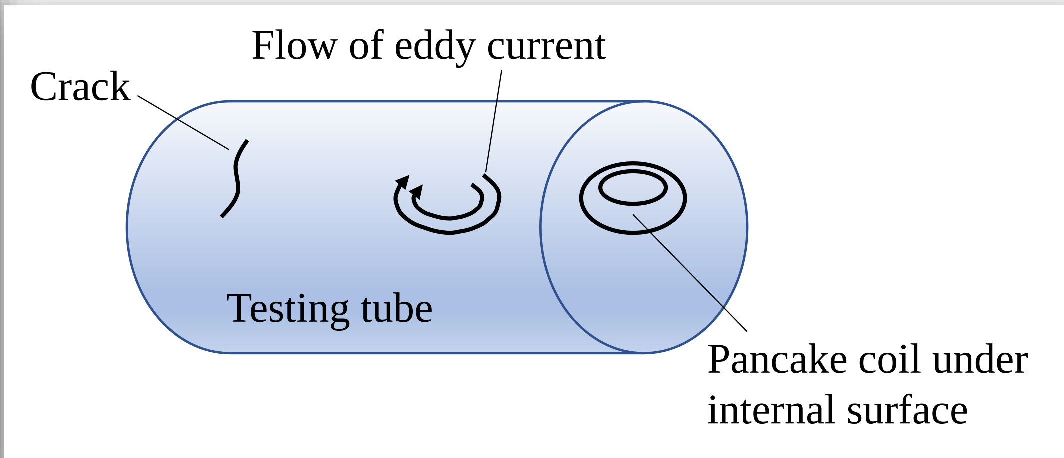
Inductive Sensors

7.6 Applications of the Inductive Sensor

7.6.3 Flaw detection sensors





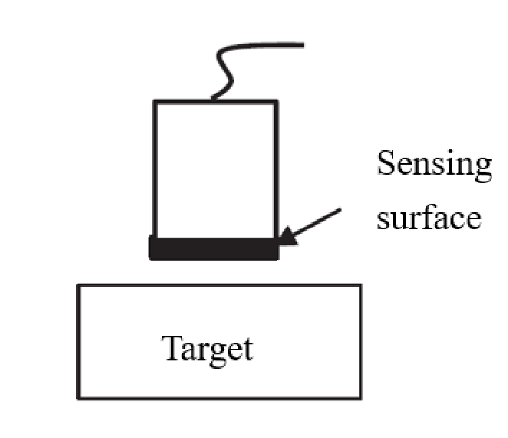


Pancake Coil Configurations on Cracked Tube

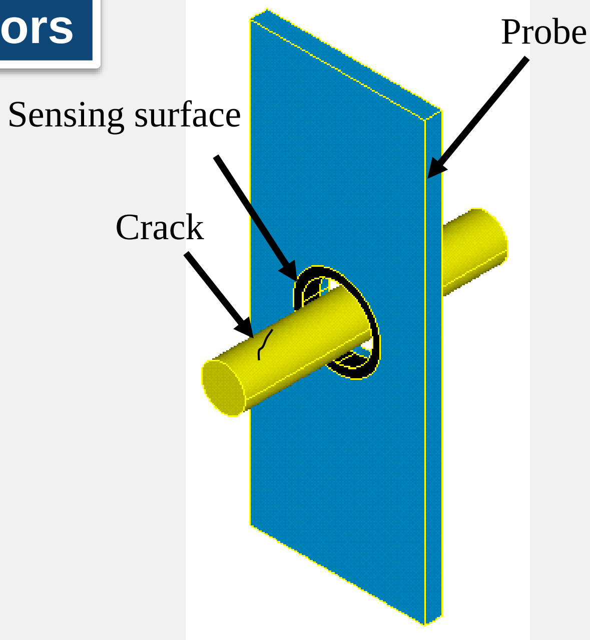
Inductive Sensors

7.6 Applications of the Inductive Sensor

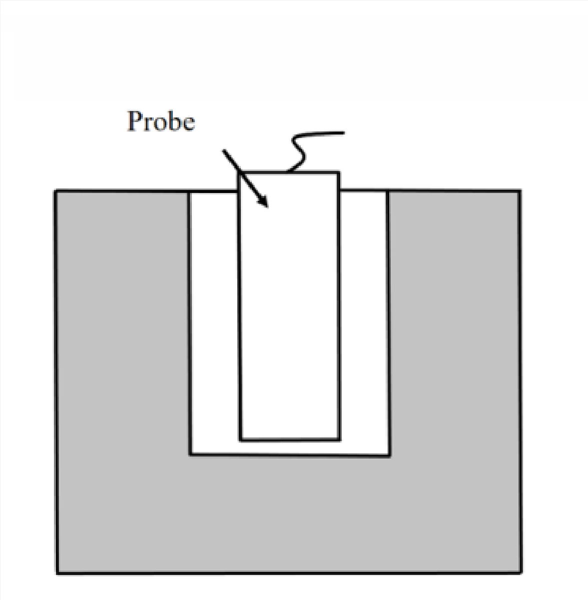
7.6.3 Flaw detection sensors



Surface probe



OD probe



ID probe

Three Types of Inductive Sensing Probes: Surface, OD, and ID

5

Summary and Consideration

三 Resistance Strain Gage

Summary and Consideration

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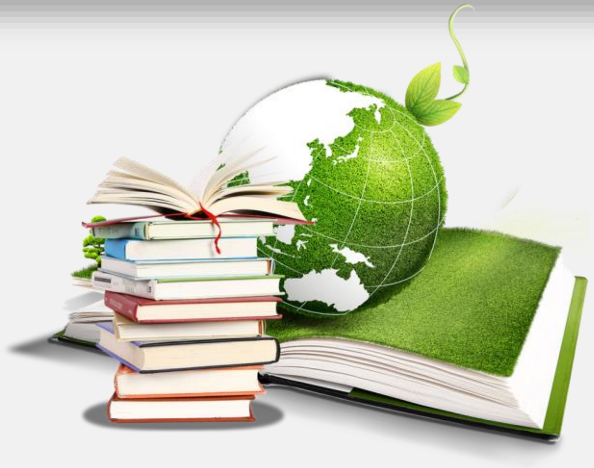
1 **Practical engineering problem**

2 **Principle**

3 **Measuring circuit**

4 **Application**

5 **Other typical application?**



Thank you .